# Gopalakrishnan Venkatesh

Incoming PhD Student in Natural Language Processing

gopuvenkat9@gmail.com gopuvenkat.github.io gopuvenkat gopuvenkat Google Scholar

## Academic Appointment

- Incoming PhD student in the ELIZA program, joining the Ubiquitous Knowledge Processing (UKP) Lab at TU Darmstadt on **15 December 2026** under the supervision of Prof. Dr. Iryna Gurevych and co-supervision of Prof. Valentin Hofmann.

## Education

**Integrated B.Tech. and M.Tech.** — **International Institute of Information Technology Bangalore (IIIT-B)**, Computer Science and Engineering
Bangalore, India
2016 – 2021
CGPA: 3.76/4.0

- Master's thesis: *Enriching Word Representations using Distributional Thesaurus*
- Teaching assistant: Operating Systems; Introduction to Automata Theory and Computability

## Experience

**Oracle**, Product Developer (Member/Senior Member of Technical Staff)
Bangalore, India
July 2021 – present
5 years 5 months

- Led AI initiatives across Essbase and Autonomous Database Data Studio, including systems that translate natural-language requests into declarative queries such as SQL.
- Contributed novel ideas and implementations that led to three patent submissions to the USPTO.
- Helped integrate Essbase with Autonomous Database by remodeling its internal calculation engine for multidimensional aggregation, calculation, and reporting.

**Google Summer of Code (GSoC) - GFOSS**, Student Developer
Remote
May 2018 – Aug 2018
4 months

- Developed **Clio**, an open-source web system for managing software component, license, intellectual-property, and related metadata.
- Added user- and group-level access control through LDAP integration.
- Mentors: Alexios Zavras and Prof. Georgia Kapitsaki

## Publications

**One Shot Learning in StyleALAE: Preserving Facial Identity During Semantic Modification** — 2022
R. K. R. Kotlo Budde, K. Shubham, **Gopalakrishnan Venkatesh**, S. Gandikota, S. Khoche, D. B. Jayagopi, G. Srinivasaraghavan
10.1145/3571600.3571617 (ICVGIP)

**Using Distributional Thesaurus to Enhance Transformer-Based Contextualized Representations for Low Resource Languages** — 2022
**Gopalakrishnan Venkatesh**, A. Jana, S. Remus, Özge Sevgili, G. Srinivasaraghavan, C. Biemann
10.1145/3477314.3507077 (ACM SAC)

**Hypernymy Detection for Low-Resource Languages: A Study for Hindi, Bengali, and Amharic** — 2022
A. Jana, **Gopalakrishnan Venkatesh**, S. M. Yimam, C. Biemann
10.1145/3490389 (ACM Transactions on Asian and Low-Resource Language Information Processing)

**Introducing Various Semantic Models for Amharic: Experimentation and Evaluation with Multiple Tasks and Datasets** 2021
S. M. Yimam, A. A. Ayele, **Gopalakrishnan Venkatesh**, I. Gashaw, C. Biemann
10.3390/fi13110275 (Future Internet)

**Learning a Deep Reinforcement Learning Policy over the Latent Space of a Pre-Trained GAN for Semantic Age Manipulation** 2021
K. Shubham, **Gopalakrishnan Venkatesh**, R. Sachdev, Akshi, D. B. Jayagopi, G. Srinivasaraghavan
10.1109/IJCNN52387.2021.9533685 (IJCNN)

**MHCAttnNet: Predicting MHC-Peptide Bindings for MHC Alleles Classes I and II Using an Attention-Based Deep Neural Model** 2020
**Gopalakrishnan Venkatesh**, A. Grover, G. Srinivasaraghavan, S. Rao
10.1093/bioinformatics/btaa479 (Bioinformatics)

**Automatic Compilation of Resources for Academic Writing and Evaluating with Informal Word Identification and Paraphrasing System** 2020
S. M. Yimam, **Gopalakrishnan Venkatesh**, J. S. Y. Lee, C. Biemann
aclanthology.org/2020.lrec-1.722 (LREC)

## Open Source Projects

**DT2ContextRep**
Code accompanying work on using a distributional thesaurus to enrich contextualized representations for low-resource languages.

**MHCAttnNet**
An attention-based neural model for predicting peptide binding across MHC class I and class II alleles.

**Clio**
An open-source web application for managing software components, licenses, intellectual property, and related metadata.

## Honors

- Conference fellowship for presenting **MHCAttnNet** at ISMB 2020.

## Teaching and Service

- Teaching assistant for **Operating Systems** and **Introduction to Automata Theory and Computability** at IIIT-B.
- Reviewer for **Sadhana - Academy Proceedings in Engineering Sciences**, published by Springer on behalf of the Indian Academy of Sciences.

## Research Interests

- Low-resource natural language processing and multilingual language technology
- Lexical semantics, representation learning, and language models
- Machine learning for structured and scientific data

## Technical Skills

**Programming:** Python, C++, C, Java, Go

**Machine Learning:** PyTorch, Keras, NumPy, Pandas

**Language Model Tooling:** LangGraph, LangChain

**Web and Research Tools:** Flask, Django, LaTeX